

Smart Crawler for Harvesting Deep web with Multi-Classification

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Abstract—In recent era data available on the internet is playing a vital role. According to research, most precious and valuable data is present in the deep web so interest in techniques to efficiently site invisible web is increasing. The challenges to extract the deep web are requirement of huge volume of resources, dynamic nature of the deep web, coverage of a wider area of deep web and higher efficiency of collected results from deep web with accuracy. Along with all the above challenges, user demand of privacy and identity is to be maintained. In this paper we proposed a smart crawler that efficiently searches the deep web and avoids visiting irrelevant pages. A smart crawler starts crawling from the center page of seed URL and goes on crawling till the last link available. The crawler has an ability to separate active and inactive links based on requests to sever of hyperlink. The crawler also contains text-based site classifier using neural network and natural language processing as Term Frequency Inverse Document Frequency and Bag of Words with supervised machine learning techniques as logistic regression, support vector machine and naive bayes. Also HTML tags are extracted from hyperlinks along with data which plays a huge role in data analysis and all this is separately saved in a centralized database. Our experimental results with efficient link reaping rate and classification show higher accuracy compared to different crawlers.

Index Terms—Deep Web, User Agent, Hyperlink, Support Vector Machine(SVM), Naive Bayes(NB), Logistic Regression(LR), Term Frequency Inverse Document Frequency(TF-IDF) and Bag of Words(BOW)

I. INTRODUCTION

The worldwide web is divided into three major parts Surface web, Deep web, and Dark web. The Surface web is part which we can be accessed through standard web browsers like Chrome, Firefox, Edge. The deep web is that part of the internet which holds the data like academic records, bank transactions, health-related information while dark web which owned an outlawed slice of information. According to research, approximately surface web incorporated 4% part while deep and dark holds the remaining 96% which is around 500-600 times of surface web [9].

The surface web is also called as Indexed web or Lightnet, this web is effortlessly available for everyone using the internet with standard browsers without any constrain, The sites like news, social media, politics, etc. lie under visible web but

data like login credentials come under deep web [5]. The deep web is part of the internet which hold un-indexable web pages, Un-indexability is because web pages are protected with credentials, robot.txt is the file of a web page which decides whether crawling will be allowed or not if yes then till what amount, some web pages have time constraint i.e. page or site will be available to access only for certain amount of time while deep web mostly contains legal portion [6][7]. The darknet is the vital part of illegal activities such as drug dealing, weapons trading, child abuse, etc. while it is a hidden network with special applications and protocols

Crawling is the process of systematically visiting several web pages to create an index of data, Crawler can also be called spiders or robots. In recent days, data has achieved a vital role in the digital era. Data extraction or data scraping has become essential however, this can be done with the help of tools like crawler's, which gives higher anonymity. Crawlers can be implemented using techniques like a focused crawler, parallel crawler, and generic crawler.

Accessing deep and dark web is not possible with standard browsers like Edge, Mozilla, and Chrome. So Tor is the browse that leads to accessing an entire deep web while Freenet and I2P are similar browsers to access the web. Tor works on the concept of onion routing in which the data of the user is first enciphered and then transmitted via different relays available inside the network to keep the user more anonymous. Also, Tor bandwidth utilization is very less compare to I2P and Freenet which makes it more selective for hidden web [8][10].

I2P is an Invisible Internet Project which works on the concept of garlic enciphering. I2P is a network layer that runs on Clearnet infrastructure; however, it uses simplex connections between every sever while also it creates virtual network into clients. Freenet is a dispersed browser that can unattested browse, publish and share files without worrying [8][9].

The deep web has an enormous amount of information available to access. The deep web has huge web pages present for accessibility which needs to capitulate queries to the database; standard crawler fails to do this and leads to raising the importance of deep web bots or deep crawlers [1][18].

Crawling data have challenges like a massive volume of resources, the dynamic nature of the web, covering a wider area and higher efficiency. The hidden databases are not certified with any traditional web engines. In contrast, these databases are distributed and keep varying continuously. So in such scenarios, a new technique mentioned in the paper can be used with higher anonymity and accuracy. As deep web crawler has multiple challenges shown in recent research such as downloading pages from the invisible web, efficient automatic link extraction, link-based classification. The classification has been given the highest priority in various domains of security. Also, User-Agent, and IP spoofing are the most stringent challenges crawlers have faced.

In this paper, study of building an efficient deep web crawler is shown. The novelty of crawler is it efficiently crawled data starting from the center page and goes on crawling subpage links available on the main page and continues. The Tor is the browser used to crawl pages from the invisible web, Crawled links are stored in the database which continuously updates, Link-based classification is comprised of three important stages. Firstly, text pre-processing, features extraction, and finally, three different supervised classifiers resulting in six different classification pipelines. Anonymity plays a valuable role so IP spoofing and user agent spoofing are added in crawler while crawler also can extract HTML tags from the links which are being crawled and this HTML tag helps showing a similar page like that on the link.

II. RELATED WORK

Information in the deep web is buried. The preceding techniques and gadget proposed by researchers are to understand and integrate the deep web with ontology approach, fuzzy-based approach, Supervised learning based approach. The Christopher Olston, Marc Najork, and Chris Sherman and Gary's price study show that crawling invisible web has three vital parts citing web content, choosing relevant content from sources and extracting the content.[13].

Web Siteing – Research shows that only 2.5% of the deep web is harvested [18]. This is because the search engines are not able to include invisible web and web sites are not available for crawling while some studies show that a huge part of this web is hosted by secure networks.

Generic crawlers are mainly designed to download the web pages and directory creation of resources. However, these crawlers are not domain-oriented and try to fetch all searchable forms. [11][16].

Ontology-based crawlers download pages after downloading RDF of these pages are read and ontology is made while hidden web miner helps to create ontology-based forms and this form is matched with ontology database if yes then hidden miner creates query and save link to the database.

Focused crawler firstly executes site-based searching and will assign a ranking to relevant sites. Further, the crawler achieves in-site searching using adaptive link ranking approach. However, crawled data may contain some duplicate

URLs or contents. These URLs can be eliminated with the help of technology like duster. [7]

Choosing relevant source – Available hidden web has very less amount of coverage for exact databases that constrain data access [12], focused crawlers are designed to crawl domain-specific content and reduce visiting off-link [6][17]. The best-first crawler uses page classification technique which only classifies whether a page is relevant or not while harvest rate is even very less so improvement to best-first crawler is proposed where a page is not only categories but the specific domain is also specified with higher harvest rate.

Feng Zhao et.al in [1] manifest that the invisible web is very huge and speedy while the dynamic nature of deep web gives a halt to wide coverage and efficiency. To overwhelm this downside paper proposes efficient crawler. Crawler allows user to give a depth of crawling i.e. it will crawl domain-specific that only number of links user want or depth given by user and if depth not given it will crawl till the last link available for crawl this will even save resources and time, so crawler can efficiently and accurately reach will wider coverage

Extraction Content from Deep Web – Most of the deep web is not easily available for surfing and needs some extraordinary techniques to access. At the same time, the invisible web also has some pages which are only available for certain timing especially drugs or weapons-related sites so crawling this web pages at off times will be waste of resource and may get false information. A proposed smart crawler can crawl the data effectively with better depth and has an ability to separate active and inactive links which increase accuracy and efficiency with a reduction of utilization resources unnecessarily [3].

A. MOTIVATION

The invisible web is inaccessible to almost all standard search engines and bots and most of the users are unaware of this vast web, As its difficult by traditional techniques to index this web. The deep web is enormous is size and contains huge amount of vital data freely available, Data may contains illegal things like child pornography, drugs dealing, money laundering, etc. which can be limit by crawling.

III. SYSTEM ARCHITECTURE

Invisible web is that part of the internet which is not available to browsing by standard web browsers like Chrome, Firefox, Edge, etc. Tor is the browser that is used to access the deep and dark web. Tor, an onion router works on circuit-based low latency overlay network that gives higher privacy and anonymity

Tor browser works on the network shown in Fig.1 which consist of Tor relays, Tor relays are also called as nodes or routers. There are three types of relays used in browser entry, middle and exit relay. The message is enciphered and send to nodes which are known, A header is peeled and routing instruction to the next router is provided whereas the message being pass is anonymous to nodes. Entry and Middle relay receive traffic entry node is where real IP of a client trying

to connect is seen, the middle relay has a vital role in the transmission and cover most of the network every node in the middle relay can act as a guard and exit node i.e. for previous nodes a current node is exit node and for next node it is entry node while no node can remember previous nodes except current guard and exit node while exit relays are relays which step back to destination sever this relay shows final IP address. Along with all these nodes, there is one more node called a bridge node that is not available in the public directory of Tor and is used to block all publicly available nodes.

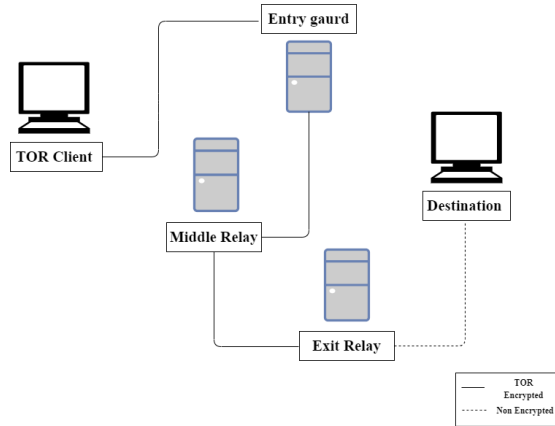


Fig. 1. Tor Network

The proposed crawler consists of sections as connection setup, seed URL with depth, data extractions, HTML tags extractions separating active and inactive links and save to database. The first section leads connection to .onion sites where proxies are used to surf the web anonymously. Here, the network IP is hidden, and proxy server IP is shown. Delay unit is added between every request to each site to avoid attacks issue like DDOS. After that connection Tor browser is set up for fetching invisible web and current IP of user is shown on a display of crawler as shown in Fig.3 whereas along with IP user agent is also visible, User-Agent for this fake user agent module of python is used in which random fake user-agent are generated which leads to higher anonymity all this section make crawler more anonymous along with this IP is spoofed using IP spoofing script which tends to more anonymity, IP shown on screen is passed through the entire process of anonymity as explain in Tor network section and last destination IP is shown. After setting up all now time to take seed URL from the user in this section user enter URL of domain interested to crawl along with depth or number of links user wants from a domain which reduces uses of resources and also saves time.

Data Fetcher or Crawling section come into picture here after entering seed URL, Crawling starts from the center page given by user in form of link however crawler begins fetching first link after complete extraction of that link it checks whether a link is active or not which is done by sending an HTTP request to link server if acknowledgment comes which decides link is active and if site failed to give

acknowledgment is leads to results in inactive link. This is shown by status called Page has found or Page has not found if found then link is active else inactive. The proposed crawler uses breadth-first search technique in which crawler first crawl current page while crawling first page it get more sub links but it will first complete current page and then it goes to sub-link from first page and process continues till the last link arrives. Domain specific link is available to extract then crawler starts from there and fetched complete current page and move to the categories if available on the same page and goes on crawling all categories present on-page and then moves to next page. .

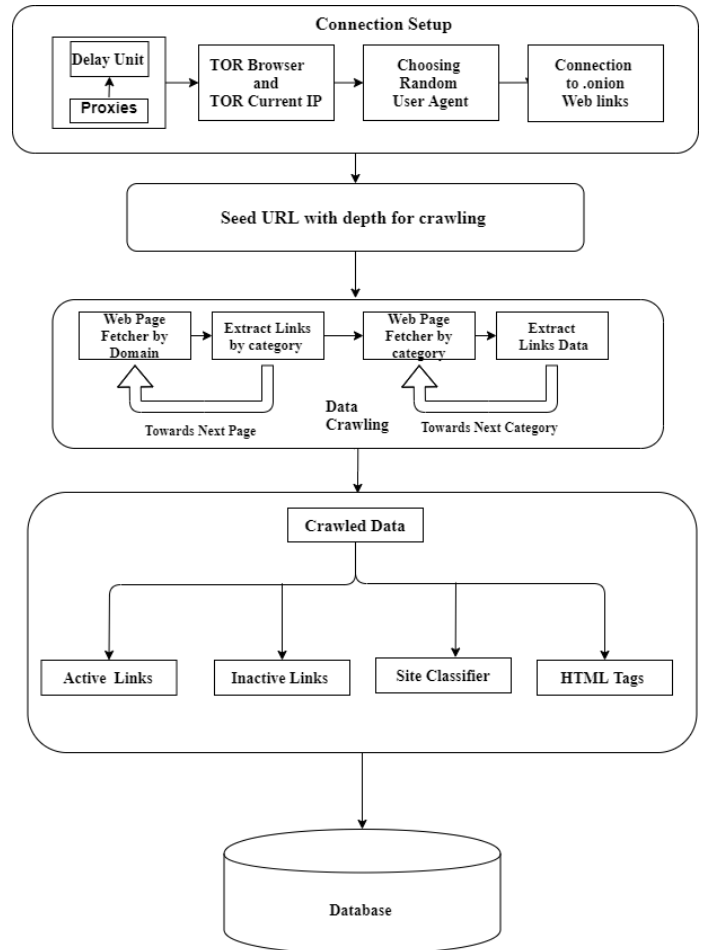


Fig. 2. Crawler Architecture

Once the links are crawled and saved into database, Active and inactive like are segregated depending on an acknowledgment of request from site then after Site Classifier comes into a role in which sites are classified based text with help of the neural network, In proposed crawler, we used Python3 with Scikit-Learn machine learning library for the pipeline's implementation. text preprocessing task is done first and after that Naive Bayes Classifier which gives good baseline, while linear support vector machine algorithm gives one of the best for text classification whereas logistic regression is utilized which can effectively categories multiple classes text in the link are

tokenized only on the links column of the dataset and applied word vector averaging to tokenized text. Classification pipeline is comprised of three main stages, First, text pre-processing, then, features extraction, and lastly classification. Proposed Crawler uses two famous text representation techniques across three different supervised classifiers resulting in six different classification pipelines, however scrutinizing every pipeline to figure out the best combination with the best parameters that can achieve high accuracy is done. Two efficient text representation technique are used first is BOW, BOW (Bag Of Words) is model that fetch features from link by counting word frequency and second is Term Frequency Inverse Document Frequency model (TF-IDF) uses a statistical model to give weight to the content of link along with these two we used three different supervised machine learning techniques Support Vector Machine, Naïve Bayes and Logistic regression are the classifiers selected and compared, Classification results are shown in section IV of the papers. HTML tags are extracted from the links which are crawled it helps in creating similar documents as that in a database which will make the user more comfortable to see the web page as it is in the database as as shown in Fig.4 while the chance of neglecting important information related to domain gets reduced which increase crawler efficiency and accuracy.

IV. RESULTS AND EXPERIMENTAL SETUP

We have design smart crawler in python3 and evaluated approach over 9 different domains as shown in Table 1. and around 2500 data samples are taken for research, Setup has several stages with python3

Domain	Description
Political	Politics of different areas in world
Forums	Few types of Forums
Hosting	Search related to Hosting
Blogs	Blogs of music news etc.
Non-English	Some Spanish, German etc.
Financial	Financial sector of different countries
Email	Different mail box available in invisible web
Marketplace	Dealing of drugs, weapons etc.
Erotic	Child abuse, Women Abuse etc.

- 1) The direct IP address to a local host which is redirected using HTTP protocol and proxies through Tor and delay of two seconds between sites is added
- 2) Enter the seed URL and start crawling through a center page and goes on crawling sublinks of the main hyperlink and data collection.
- 3) Separate Active and Inactive links based on a request to server of sites
- 4) Site Classification using machine learning and neural networks with comparison of different supervised learning algorithm.
- 5) HTML tags extraction from the links and creating similar pages as that of site.
- 6) Extracted data will be segregated and saved into specific section of database.

The figure 3 shows how the crawler efficiently works with .onion sites i.e. deep web while above data extraction is shown on the financial site of the invisible web whereas the user can also see the IP address after all spoofing and going through Tor network. As in above section, we talked about the anonymity of this proposed crawler here are results of that where the user agent is visible and even either page is found or not is also shown which differentiate hyperlinks into active or inactive i.e. if a page is found which leads to decide link is active if a page is not found then will lead as an inactive page.

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Your Current IP is: 31.228.2.133
Currently you are on site 1 : http://kkg2pmhq6b4y5xu.onion/channel/UCnZdV89VFyJrLbU1Y59KxaQ
Page has found....
User Agent is : Mozilla/5.0 (Macintosh; Intel Mac OS X 10_10_1) AppleWebKit/537.36 (KHTML, like
1 http://kkg2pmhq6b4y5xu.onion/
2 http://kkg2pmhq6b4y5xu.onion/toggle_theme?referer=k2Fchannelk2FUCnZdV89VFyJrLbU1Y59KxaQ
3 http://kkg2pmhq6b4y5xu.onion/preferences?referer=k2Fchannelk2FUCnZdV89VFyJrLbU1Y59KxaQ
4 http://kkg2pmhq6b4y5xu.onion/Login?referer=k2Fchannelk2FUCnZdV89VFyJrLbU1Y59KxaQ
5 http://kkg2pmhq6b4y5xu.onion/feed/channel/UCnZdV89VFyJrLbU1Y59KxaQ
6 http://kkg2pmhq6b4y5xu.onion/Login?referer=k2Fchannelk2FUCnZdV89VFyJrLbU1Y59KxaQ
7 https://www.youtube.com/channel/UCnZdV89VFyJrLbU1Y59KxaQ
8 http://kkg2pmhq6b4y5xu.onion/channel/UCnZdV89VFyJrLbU1Y59KxaQ/playlists
9 http://kkg2pmhq6b4y5xu.onion/channel/UCnZdV89VFyJrLbU1Y59KxaQ/community
10 http://kkg2pmhq6b4y5xu.onion/channel/UCnZdV89VFyJrLbU1Y59KxaQ?page=1&sort_by=oldest
11 http://kkg2pmhq6b4y5xu.onion/channel/UCnZdV89VFyJrLbU1Y59KxaQ?page=1&sort_by=popular
12 http://kkg2pmhq6b4y5xu.onion/watch?v=H_9rh0UwFs
13 http://kkg2pmhq6b4y5xu.onion/watch?v=H_9rh0UwFs
14 http://kkg2pmhq6b4y5xu.onion/channel/UCnZdV89VFyJrLbU1Y59KxaQ

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Fig. 3. Data Crawling Section

Fig.4 shows HTML tags helping to rebuild a similar page as that of the site so which reduce efforts and even use of resources, however this data from HTML tags can be use in applications like data analysis, data management, etc.

Newly updated List - Deep Web Links 2017, .onion sites 2017, Deep Web linkert, for Links, Dark Websites, Deep web websites, Deep Onion sites, the deep web, dark Internet. The web is the hidden part of the internet which is not indexed or crawled by any standard search engine like Google or Yahoo. No idea how big it is? And you might not have visited any deep web links that are hidden in the dark world. According to 4% of the web is visible to the public and the rest of the 96% are hidden underneath. The links available on the deep web are not available in the normal search engines. You can get them here. Related: 9 things you didn't know about the deep web. It is one of the best Deep/Dark Web Links which is been updated today. You can visit those links here. Before accessing, I strongly recommend you to use a VPN. This could prevent you from being tracked or watched by any hacker. Be anonymous. An expert team is keen on finding New Deep Web Links which are found to be interesting. These deep web links is a great tool on such as for discovering deep websites and further go down into deep web research, follow the security tips recommended. Security tips for deep web users I know we are all curious in finding what is in the deep web but there have been many thing the cautions and security problems that might arise if you use them. Have you ever heard some of the deep web facts? The place where anonymity is the key factor. Security tips: to remain anonymous, The Deep Web team strongly recommends you surf the deep web safely. Also, compare other Powerful VPN Service too. Be safe. For one to stay anonymous he must do it right. First, you must install the Tor Browser (Step-by-Step Instruction guidelines are given). To prevent the theft of your personal data breach we strongly recommend you to Setup a VPN to hide your IP or buy these Best VPN Directly. Hidden Wiki List is to start for the newbies, get to know how to register and purchase from various darknet markets. In order to purchase if you need to buy bitcoins from top bitcoin vendors. "How to find deep web links, what is deep web links, what is called as where to find deep web links." This is the questions that will arise in the mind of everyone so here we present the solution. own this path know about what is the deep web. Read more about >>> Darknet <<< here. Table of Content: Top 50 Dark Web Links (Latest Update 2017) .onion links 2017 (Deep Web) Forums/Chan and Community Links (Deep web links Url) Anonymous Web Links Bitcoin, Money, Paypal Accounts, Credit Cards and Financial Service Deep Web Links Forums, Chans, Hosting and (Updated) Deep web Drugs Markets/Store Links (Updated 2017) Updated Dark Web Links (Deep web Search Engine) Deep web Links (Commercial) Best Wikis, Forums, Search Engines and Blogs Best Darknet Sites and Markets HIDDEN WIKI - DEEP WEB UNDERGROUND Latest Update List : Deep web Links 2017, The hidden wiki, Deep web sites, Dark web Search, The Dark Web Link for hidden wiki links, deep web sites links, links deep web sites 2017, Tor links, dark web sites, links da deep web 2017, darknet links 2017, uncensored hidden wiki, .onion links, deep web link 2017 and tor directory. Access the hidden deep web. We have listed out some of the rarest links from the dark world. You might not see some of the links on any

Fig. 4. HTML tags extracted and similar document created

We have categorized links of 9 different domains using TF-IDF and BOW with LR, NB and SVM to test and train we created a dataset of around 2500 links and data of nine classes every class contain a recall, a precision and an F1 score. To combine these three values into a single value, We calculated the weighted average, macro and micro for each class, to train classifiers firstly we transformed tags column in a dataset into a vector of number as TF-IDF weighted vector after that we train data set with these three classifiers

and tested, Along with all this we have used Keras an neural network API with TensorFlow backend which shapes input data, output data and make predication of link more easier with higher accuracy. Above Figure shows that combination of TF-IDF along with Logistic regression classifier got 94.44% of accuracy and 94% macro F1 score, whereas SVM got 92.42% and NB with 88.88% accuracy but as use of neural network in classification achieves higher accuracy with LR of 95.46% Logistic regression with TF-IDF and neural network has achieved higher accuracy than Support Vector Machine and Naïve Bayes and made prediction of link more efficient and accurate in multi-classifying however dataset can be further extended and made wider to increase accuracy.

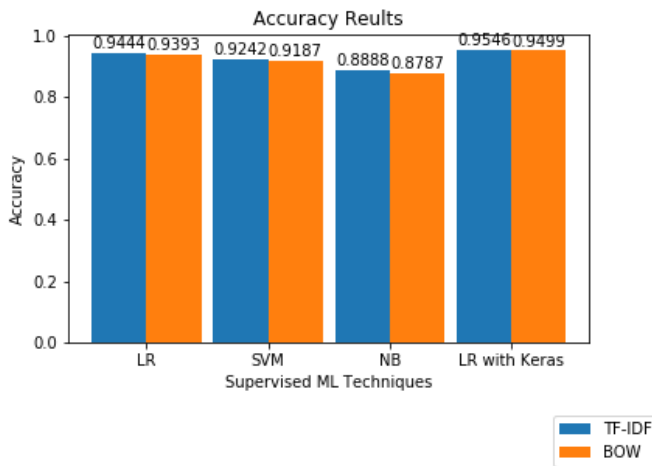


Fig. 5. Comparison of accuracy with all three supervised learning technique

V. CONCLUSION

In this paper we proposed a smart crawler that efficiently searches the deep web and avoids waste of resources and time, A smart crawler starts crawling from the center page and goes on crawling till the depth given or elsewhere till the last link available. The crawler also has an ability to separate active and inactive links by requesting to web sever of sites, Whereas crawler also contains text-based site classifier using a neural network with supervised machine learning techniques in which we achieved best 95.46% of accuracy in classifying sites while machine learning algorithm Logistic Regression with TF-IDF has got accuracy F1-score of 94%, Increment in accuracy can be obtain by further improvement in dataset. Accompanying HTML tags extraction from sites gives more preferable and valuable data that can make the processing, analysis, management, etc. more comfortable, easy and time saving whereas all this is separately saved in a centralized database. Our experimental results show sharpness and accuracy of our proposed crawler which effectively achieves higher reaping rates compared to different crawlers, In the feature, we will try to increase size of dataset and we will be looking for keyword-based crawling, Along with Tor will like to explore

I2P and Freenet and comparison of these three and their ability to work with the invisible web.

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